

Cardiomyopathy Dilated 2H (CMD2H) — Comprehensive Disease Characterization Report

Disease: Cardiomyopathy, dilated, 2H (CMD2H) **OMIM:** #620203 · **MONDO:** MONDO:0859358 · **MedGen/UMLS:** C5774296 · **GARD:** 0026714 **Causal gene:** *RPL3L* (HGNC:10351; UniProt Q92901) · **Category:** Mendelian (autosomal recessive)

Summary

Cardiomyopathy dilated 2H (CMD2H; OMIM #620203; MONDO:0859358) is a rare, **autosomal-recessive, neonatal-onset dilated cardiomyopathy (DCM)** caused by biallelic pathogenic variants in *RPL3L* (ribosomal protein L3-like; HGNC:10351; NCBI Gene 6123; UniProt Q92901). *RPL3L* encodes the **heart- and skeletal-muscle-specific paralog of the core large-ribosomal-subunit protein RPL3 (uL3)** — a striking example of tissue-specialized ribosome biology underlying a Mendelian cardiac disease. The disease is defined at the disease level in aggregated ontology/genomic resources (OMIM, MONDO, ClinVar, gnomAD) and clinically from a small number of individual patient case series identified by whole-exome/whole-genome sequencing (fewer than ~15 families worldwide).

Mechanistically, CMD2H is best explained by a **combined loss-of-function/gain-of-function** model. In healthy striated muscle, *RPL3L* and its ubiquitous paralog *RPL3* exist in a dynamic balance in which *RPL3L* represses *RPL3* (in part via unproductive splicing). *RPL3L* is loss-of-function-tolerant in the general population (gnomAD pLI $\approx$ 0; observed/expected LoF $\approx$ 0.99; LOEUF $\approx$ 1.27), and simple loss of *RPL3L* is buffered by compensatory *RPL3* upregulation — the reason *Rpl3l*-knockout mice have no overt cardiac phenotype. Severe human disease therefore requires an allele that not only removes *RPL3L* function but also **blocks the protective RPL3 compensation**. Recurrent **hotspot missense variants** accomplish this by inducing **nucleolar protein aggregation, disrupting rRNA processing, and preventing RPL3 upregulation**, thereby crippling muscle-specific ribosome biogenesis in cardiomyocytes.

Clinically, CMD2H presents as **rapidly progressive, high-mortality neonatal heart failure**. Management is entirely supportive and mirrors severe pediatric DCM: guideline-directed heart-failure pharmacotherapy, mechanical circulatory support (ECMO and ventricular assist devices), and cardiac transplantation. No gene-specific or disease-modifying therapy exists. Because affected infants are among the youngest and most fragile in the pediatric cardiomyopathy population, outcomes without transplantation are poor, though modern VAD/transplant pathways achieve excellent survival once a child is successfully bridged.

Key Findings

Finding 1 — Biallelic *RPL3L* variants cause CMD2H (autosomal recessive neonatal DCM)

Whole-exome sequencing established *RPL3L* as the causal gene for a distinct, neonatal-onset DCM. In the index study, compound heterozygous missense variants co-segregated with severe neonatal DCM in **five affected individuals from three independent families**: *"we identified causative, compound heterozygous missense variants in RPL3L (ribosomal protein L3-like) in all the affected individuals"* [PMID: 32514796](#). A second, independent family confirmed autosomal-recessive inheritance, carrying a maternal frameshift allele (c.1076_1080delCCGTG, p.Ala359Glyfs4) in trans with a paternal missense allele (c.80G>A, p.Gly27Asp); that report concluded *"our findings support the pathogenicity of biallelic RPL3L pathologic variants associated with rapidly progressive neonatal DCM and heart failure with a poor prognosis"* [PMID: 35323613](#). Causal alleles are absent or very rare in population databases, consistent with a recessive Mendelian disorder. (Evidence source: human clinical/genetic.)*

Finding 2 — Dual loss-of-function / gain-of-function mechanism blocks protective RPL3 compensation

Work integrating human genetics, patient tissues, and isogenic cell models refined the mechanism. Affected individuals typically carry **one of two recurrent hotspot missense variants paired with a private (family-specific) allele**. Critically, non-hotspot variants phenocopy a knockout and permit compensatory RPL3 upregulation, whereas hotspot variants add a toxic gain of function: *"hotspot variants induce nucleolar protein aggregation, disrupt rRNA processing and block compensation by preserving the role of RPL3L in repressing RPL3 via unproductive splicing"* [PMID: 41495453](#) (see also the mechanistic review, [PMID: 39803500](#)). The

compensatory axis is corroborated in mice: *Rpl3l* knockout produces no overt cardiac phenotype, and *"depletion of RPL3L leads to increased ribosome-mitochondria interactions in cardiomyocytes, which is accompanied by a significant increase in ATP levels"* PMID: 36882085. This establishes the RPL3/RPL3L balance as a regulator of cardiac mitochondrial activity and explains why a compensation-blocking allele is required for human disease. (Evidence sources: human genetic, in vitro isogenic models, model organism.)

Finding 3 — *RPL3L* coding variation links the gene to cardiac electrophysiology (atrial fibrillation)

Beyond the recessive cardiomyopathy syndrome, common and low-frequency *RPL3L* coding variants associate with **atrial fibrillation (AF)**. A large GWAS meta-analysis (29,502 AF cases; 767,760 controls; Iceland + UK Biobank with replication), *"focusing on low-frequency coding and splice variants aiming to identify causal genes,"* implicated *RPL3L*, including a missense association (OR $\approx$ 1.20) and splice-variant signals PMID: 30271950. *RPL3L* also appears among loci associated with **P-wave duration**, an atrial-conduction endophenotype shared with AF PMID: 32822252. This positions *RPL3L* as a heart-specific gene with a phenotype spectrum broader than CMD2H alone. (Evidence source: human population genetics.)

Finding 4 — Pediatric DCM epidemiological and prognostic context

CMD2H sits within a rare and severe disease space. Population-based pediatric cardiomyopathy incidence is approximately **0.9 per 100,000 children per year** (Northern Ireland national cohort, 2003–2025; 84 children): *"corresponding to a mean annual incidence of 0.9 per 100,000 children"* PMID: 42360425. In the ESC EURObservational childhood cardiomyopathy registry, **DCM comprised 206/633 (32.5%)** cases (*"dilated [DCM; n = 206 (32.5%)]"*), with a median age at diagnosis of 4 years, a pathogenic/likely-pathogenic variant in 60.4% of genetically tested patients, and ~3% 1-year mortality PMID: 38427064. Prognostically, plasma NT-proBNP is a validated biomarker: *"plasma NT-proBNP concentrations predicted the risk of heart transplantation or death in children with HF"* PMID: 38722325. Contemporary end-stage management relies on VAD (HeartMate 3) and transplantation, with excellent 3-year survival (~94–96%) once supported PMID: 42334151. (Evidence source: human clinical/registry.)

Finding 5 — *RPL3L* is loss-of-function tolerant in gnomAD (supports the compensation-blocking requirement)

Population constraint metrics reinforce the mechanistic model. For *RPL3L* (NCBI Gene 6123; HGNC:10351; ENSG00000140986; chr16:1,943,791–1,957,606, GRCh38, minus strand), gnomAD reports **pLI = 2.3×10^{-15} (≈ 0)**, **observed/expected LoF = 0.99 (90% CI 0.78–1.27)** (LOEUF ≈ 1.27), and **LoF $z \approx 0.06$** — i.e., no depletion of loss-of-function variants. Simple haploinsufficiency is therefore unlikely to cause disease, consistent with the requirement for a gain-of-function/compensation-blocking allele. ClinVar lists **200 *RPL3L* variants, 62 classified pathogenic/likely-pathogenic** (canonical RefSeq NM_005061.3), spanning nonsense (p.Gln175Ter, p.Trp257Ter, p.Arg97Ter, p.Gln147Ter), frameshift (p.Thr340fs), splice (c.849+5G>A), and missense (p.Pro82Ala, p.Arg100Trp, p.Gly270Arg, p.Thr346Met, p.Glu384Lys) classes. The protein is UniProt Q92901, 407 amino acids. (*Evidence source: computational/population database.*)

Finding 6 — Authoritative identifiers resolved

The disease is uniquely identified as **OMIM #620203 / MONDO:0859358** (EBI OLS4 returns MONDO:0859358 with exact label "cardiomyopathy, dilated, 2H"), correcting an earlier working assumption of #619371. Equivalent cross-references: **OMIM:620203, MedGen/UMLS C5774296 (MedGen UID 1824069), GARD:0026714**. No CMD2H-specific Orphanet code exists (grouped under ORPHA:154, familial isolated DCM). MONDO:0859358 is a child of MONDO:0005021 (dilated cardiomyopathy). Gene identifiers: *RPL3L* OMIM 617416, HGNC:10351, NCBI Gene 6123, ENSG00000140986, RefSeq NM_005061.3, UniProt Q92901.

Finding 7 — Functional and ortholog annotations finalized

Verified GO annotations for *RPL3L*: **BP** — GO:0002181 (cytoplasmic translation), GO:0006412 (translation), GO:0006941 (striated muscle contraction), GO:0010832 (negative regulation of myotube differentiation), GO:0016202 (regulation of striated muscle tissue development); **MF** — GO:0003735 (structural constituent of ribosome), GO:0003723 (RNA binding); **CC** — GO:0022625 (cytosolic large ribosomal subunit), GO:0005840 (ribosome). Mouse ortholog *Rpl3l* = NCBI Gene 66211, MGI:1913461, ENSMUSG00000002500, chromosome 17. Relevant cell type: CL:0000746 (cardiomyocyte); anatomy: UBERON:0000948 (heart), UBERON:0002084 (heart left ventricle).

Report by Template Section

1. Disease Information

Overview. CMD2H is a Mendelian, autosomal-recessive DCM with **neonatal onset** and rapidly progressive heart failure. DCM is defined by left-ventricular (or biventricular) dilatation and reduced systolic function in the absence of abnormal loading or coronary disease. CMD2H is distinguished from the broad DCM family by its very early presentation and its molecular basis in a **muscle-specific ribosomal protein paralog** rather than the sarcomeric/cytoskeletal genes that dominate DCM genetics [PMID: 32514796](#); [PMID: 35323613](#).

Resource	Identifier
OMIM (disease)	#620203
MONDO	MONDO:0859358 ("cardiomyopathy, dilated, 2H")
MedGen / UMLS	C5774296 (MedGen UID 1824069)
GARD	0026714
Orphanet	No CMD2H-specific code; grouped under ORPHA:154 (familial isolated DCM)
ICD-11 / ICD-10	BC43.1 / I42.0 (Dilated cardiomyopathy)
MeSH	Cardiomyopathy, Dilated (D002311)
Parent term	MONDO:0005021 (dilated cardiomyopathy)
Causal gene (OMIM)	<i>RPL3L</i> , OMIM 617416

Synonyms. CMD2H; DCM 2H; RPL3L-related / RPL3L-associated dilated cardiomyopathy; RPL3L-associated neonatal dilated cardiomyopathy; autosomal recessive dilated cardiomyopathy 2H.

Information source. Disease-level identifiers and constraint/variant data derive from **aggregated resources** (OMIM, MONDO, ClinVar, gnomAD, GO). Clinical/phenotypic knowledge derives from **individual patient case series** identified by exome/genome sequencing (small n).

2. Etiology

Primary cause — genetic. Biallelic (homozygous or compound heterozygous) pathogenic *RPL3L* variants, clustering in the conserved RPL3 domain and absent/ultra-rare in population databases [PMID: 32514796](#). Affected individuals typically carry **one of two recurrent hotspot missense variants paired with a private allele** [PMID: 41495453](#).

Genetic risk factors. The disease-defining risk is two damaging *RPL3L* alleles; heterozygous carriers are unaffected (recessive) [PMID: 35323613](#). A separate **common/low-frequency variant axis** links *RPL3L* to atrial fibrillation and P-wave duration [PMID: 30271950](#); [PMID: 32822252](#).

Environmental risk factors. None established as causal. General DCM modifiers (perinatal stress, intercurrent infection, arrhythmia) may influence timing/severity of decompensation but are not disease-initiating.

Protective factors. The key biological protection is **compensatory RPL3 upregulation**, which rescues loss of RPL3L; hotspot alleles defeat this intrinsic mechanism [PMID: 41495453](#); [PMID: 36882085](#). No environmental protective factors are defined.

Gene–environment interactions. Not characterized. The dominant interaction is **gene–gene (RPL3L–RPL3 paralog compensation)** rather than gene–environment.

3. Phenotypes

CMD2H presents in the **neonatal period / early infancy** with rapidly progressive heart failure.

Phenotype	Type	HPO term	Onset / severity / frequency
Dilated cardiomyopathy	Clinical/imaging sign	HP:0001644	Neonatal; severe; defining (~universal)
LV systolic dysfunction / reduced EF	Lab/imaging	HP:0005162	Neonatal; severe; typical
Congestive heart failure	Clinical sign	HP:0001635	Neonatal–infantile; severe; frequent
Cardiogenic shock / poor perfusion	Clinical sign	HP:0100750	Acute; severe
Tachypnea / respiratory distress	Symptom	HP:0002789	Early; common
Poor feeding / failure to thrive	Symptom	HP:0011968 / HP:0001508	Infantile; common
Cardiomegaly	Imaging sign	HP:0001640	Neonatal; frequent
Arrhythmia (incl. atrial)	Clinical sign	HP:0011675	Variable
Elevated NT-proBNP	Lab abnormality	HP:0031185	Neonatal; prognostic PMID: 38722325
Endocardial fibroelastosis (some)	Pathology	HP:0004301	Variable

Characteristics. Onset neonatal/congenital; severity severe; course rapidly progressive to end-stage heart failure; DCM and systolic dysfunction essentially universal, remaining signs being heart-failure sequelae. **Quality of life:** profound impact — intensive-care dependence, feeding/growth failure, need for mechanical support or transplantation. Standardized QOL instruments are not reported for this ultra-rare condition.

4. Genetic / Molecular Information

- **Causal gene:** *RPL3L* (60S ribosomal protein L3-like; UniProt Q92901, 407 aa; chr16p13.3). Encodes the paralog that **replaces RPL3 (uL3) in the ribosomes of striated cardiac/skeletal muscle** [PMID: 32514796](#); [PMID: 36882085](#).

- **Variant landscape (ClinVar, NM_005061.3):** 200 variants, **62 pathogenic/likely-pathogenic** — nonsense (p.Gln175Ter, p.Trp257Ter, p.Arg97Ter, p.Gln147Ter), frameshift (p.Thr340fs, p.Ala359Glyfs4), *splice* (c.849+5G>A), *missense* (p.Pro82Ala, p.Arg100Trp, p.Gly270Arg, p.Thr346Met, p.Glu384Lys). Two recurrent hotspot missense* alleles feature prominently [PMID: 41495453](#).
- **Allele frequency / constraint:** causal alleles absent/very rare; overall the gene is **LoF-tolerant** (gnomAD pLI $\approx$ 0; o/e LoF = 0.99; LOEUF $\approx$ 1.27) — consistent with RPL3 buffering of simple loss.
- **Origin:** germline, inherited in trans from unaffected carriers.
- **Functional consequence: combined LoF + GoF** — non-hotspot alleles = LoF (phenocopy knockout, permit RPL3 compensation); hotspot alleles add toxic aggregation and block compensation [PMID: 41495453](#).
- **Modifier gene:** *RPL3* — its upregulation capacity determines allele tolerability [PMID: 36882085](#).
- **Epigenetics / chromosomal abnormalities:** none disease-specific; single-gene SNV/indel disorder.

5. Environmental Information

No environmental factors, lifestyle factors, or infectious agents are established as causes or triggers (monogenic disease). Intercurrent viral illness may precipitate decompensation but does not cause disease; **viral myocarditis is a key differential** to exclude.

6. Mechanism / Pathophysiology

Ordered causal chain (initiating lesion → clinical manifestation):

1. **Biallelic *RPL3L* genotype** (typically one recurrent hotspot missense allele + one private allele) is present in cardiomyocytes → *demonstrated (human genetics)*.
2. **Leads to** defective RPL3L in the muscle-specific ribosome, where RPL3L normally substitutes for RPL3 (uL3) → *demonstrated*.
3. **Branch A (non-hotspot/LoF alleles):** RPL3L loss **results in** compensatory **RPL3 upregulation**, restoring a functional ribosome (milder/rescued path; phenocopies the benign KO mouse) → *demonstrated (cell + mouse; PMID: 36882085; PMID: 41495453)*.
4. **Branch B (hotspot/GoF alleles):** mutant protein **leads to nucleolar aggregation** and **disrupts rRNA processing**, while **preserving RPL3L repression of RPL3** (unproductive

splicing), which **blocks compensatory RPL3 upregulation** → *demonstrated (isogenic cells; PMID: 41495453)*.

5. Blocked compensation + disrupted ribosome biogenesis **result in** deficient/aberrant muscle-specific translation → *inferred/demonstrated*.
6. Perturbed ribosome pools **alter ribosome-mitochondria interactions and cardiac energetics (ATP)** → *demonstrated in mouse (PMID: 36882085); human contribution inferred*.
7. Impaired proteostasis/energetics **lead to** contractile failure and remodeling → **chamber dilatation, reduced systolic function (DCM)** → *inferred from phenotype*.
8. Progressive pump failure **results in** congestive heart failure, cardiogenic shock, and untreated neonatal death → *demonstrated clinically (PMID: 35323613)*.

Population-genetic corroboration. RPL3L's LoF-tolerance in gnomAD independently supports the branch structure: pure loss is buffered (Branch A), so severe disease preferentially requires the **compensation-blocking GoF** allele class (Branch B) — consistent with the benign KO mouse and the biochemical hotspot data.

Category detail. *Molecular pathways:* ribosome biogenesis / rRNA processing (nucleolus), mRNA translation, RPL3 autoregulation via unproductive splicing (KEGG "Ribosome" map03010; Reactome rRNA processing/translation). *Cellular processes:* nucleolar protein aggregation, proteostasis stress, mitochondrial regulation, cardiomyocyte contractile dysfunction. *Protein dysfunction:* GoF aggregation (hotspot) / LoF (others). *Metabolic changes:* altered mitochondrial activity/ATP PMID: 36882085. *Immune involvement:* none primary. *Suggested ontology terms:* BP GO:0002181, GO:0006412, GO:0006941, GO:0010832, GO:0016202, GO:0042254, GO:0006364; MF GO:0003735, GO:0003723; CC GO:0022625, GO:0005840, GO:0005730; cell type CL:0000746.

7. Anatomical Structures Affected

- **Organ level:** primary — **heart** (UBERON:0000948), especially **left ventricle** (UBERON:0002084) and myocardium (UBERON:0002349). Secondary — lungs (pulmonary congestion), liver (congestive hepatopathy), systemic hypoperfusion. Body system: **cardiovascular** (with respiratory involvement).
- **Tissue/cell level:** cardiac muscle tissue; target cell = **cardiomyocyte (CL:0000746)**; skeletal muscle expresses RPL3L but clinical involvement is cardiac-predominant.
- **Subcellular level:** **nucleolus** (GO:0005730; aggregation site), ribosome/cytosol (GO:0005840, GO:0022625), mitochondrion (GO:0005739).

- **Localization / laterality:** cardiac, typically biventricular/global with LV predominance; not lateralized.

8. Temporal Development

- **Onset:** neonatal/congenital to early infancy; acute-to-subacute presentation with heart failure.
- **Progression:** rapidly progressive; moves quickly from compensated dysfunction to end-stage heart failure; course is progressive (not relapsing-remitting).
- **Duration/patterns:** without transplant, frequently lethal in infancy [PMID: 35323613](#); spontaneous remission not described; durable "remission" only via transplantation. The neonatal window is the critical period for diagnosis and intervention.

9. Inheritance and Population

- **Inheritance:** autosomal recessive; unaffected carrier parents [PMID: 32514796](#); [PMID: 35323613](#).
- **Penetrance/expressivity:** high penetrance for neonatal DCM in biallelic individuals; variable severity correlating with allele class (hotspot vs non-hotspot) [PMID: 41495453](#).
- **Founder effects/consanguinity:** recurrent hotspot alleles suggest mutational hotspots or founder contributions; consanguinity can generate homozygous cases.
- **Carrier frequency:** not established; causal alleles individually ultra-rare (gnomAD).
- **Epidemiology:** ultra-rare (<~15 families). Context: pediatric cardiomyopathy incidence $\approx$ 0.9/100,000 children/year [PMID: 42360425](#); DCM = 32.5% of pediatric cardiomyopathy in the ESC registry [PMID: 38427064](#).
- **Sex ratio / geography:** no strong sex bias described (recessive, neonatal); reported across European and other populations without established endemic clustering. (Broader DCM/ACM populations show ~61% male predominance with gene-specific variation [PMID: 42159538](#), not established for *RPL3L*.)

10. Diagnostics

- **Imaging:** echocardiography is primary (dilated LV, reduced EF/FS); cardiac MRI adjunctive; CXR shows cardiomegaly/congestion.
- **Biomarkers:** elevated **NT-proBNP/BNP** and troponin; NT-proBNP predicts death/transplant in children with HF [PMID: 38722325](#).

- **Electrophysiology:** ECG for chamber enlargement/arrhythmia (note *RPL3L*–AF/P-wave links, [PMID: 30271950](#); [PMID: 32822252](#)).
- **Genetic testing (definitive):** WES/WGS established all reported cases [PMID: 32514796](#); *RPL3L* is in expanded cardiomyopathy/DCM panels (NM_005061.3). Interpretation caveat: because *RPL3L* is **LoF-tolerant**, an isolated truncating allele should be interpreted cautiously — the disease-relevant genotype is **biallelic**, typically pairing a hotspot missense allele with a second variant; confirm **parental segregation (in trans)**. CMA/karyotype/FISH not indicated; mtDNA and repeat-expansion testing are for differentials.
- **Biopsy/pathology:** endomyocardial biopsy shows nonspecific myocyte changes / occasional endocardial fibroelastosis; mainly used to exclude myocarditis.
- **Differential diagnosis:** viral myocarditis; metabolic/mitochondrial cardiomyopathies (e.g., **Barth syndrome**/*TAFAZZIN* [PMID: 40645388](#)); other genetic neonatal DCM (*NRAP*, *TNNI3*, *TNNT2*, *ACTC1*, *MYH7*, *LMOD2*, *ALPK3*); storage disorders [PMID: 30384889](#); [PMID: 32458740](#).
- **Screening:** cascade genetic testing; carrier/preconception, prenatal, or preimplantation testing for couples with a prior affected child.

11. Outcome / Prognosis

- **Survival/mortality:** poor — "rapidly progressive ... heart failure with a poor prognosis" [PMID: 35323613](#). In broader neonatal/infantile DCM cohorts, survival free from death/transplant is ~66–69% at 1 year and ~50% at 6 years, worst with onset in the first month [PMID: 18652581](#).
- **Morbidity/function:** intensive-care dependence, growth failure, complications of heart failure and mechanical support.
- **Complications:** cardiogenic shock, arrhythmia, thromboembolism (e.g., stroke on Berlin Heart EXCOR [PMID: 35849328](#)), multiorgan hypoperfusion, transplant-related morbidity.
- **Recovery potential:** myocardial recovery uncommon; definitive treatment is transplantation. Once bridged, adolescents/children with DCM achieve ~94–96% 3-year survival [PMID: 42334151](#).
- **Prognostic factors:** allele class (hotspot vs non-hotspot) [PMID: 41495453](#); earliest/severe onset [PMID: 18652581](#); NT-proBNP [PMID: 38722325](#).

12. Treatment

No gene-specific therapy; management is standard pediatric heart-failure care plus advanced support.

Modality	Examples	NCIT (suggested)
Heart-failure pharmacotherapy	Diuretics, ACE inhibitors/ARBs, beta-blockers, MRAs; anticoagulation	NCIT:C98065
Inotropic/critical-care support	Milrinone, epinephrine; ICU management	—
Mechanical circulatory support	ECMO; VAD (Berlin Heart EXCOR, HeartMate 3)	NCIT:C50076 (Ventricular Assist Device)
Definitive therapy	Orthotopic heart transplantation	NCIT:C15325 (Heart Transplantation)

Left-heart decompression (balloon atrial septostomy vs LA cannulation) is used in ECMO-supported infants [PMID: 38053305](#). VAD-to-transplant bridging is standard [PMID: 31298178](#); [PMID: 42334151](#). No approved gene/RNA therapy; the dual LoF/GoF model suggests rational future strategies — **allele-selective knockdown (ASO/siRNA) of the toxic hotspot allele** or approaches that **de-repress RPL3** [PMID: 41495453](#) (preclinical/conceptual). No established pharmacogenomics for CMD2H.

13. Prevention

- **Primary prevention:** not possible for a germline recessive disorder; **genetic counseling** (25% recurrence risk per pregnancy for carrier couples).
- **Secondary prevention:** early neonatal recognition of heart failure enables timely support/ listing; **cascade screening** of relatives.
- **Tertiary prevention:** guideline-directed heart-failure therapy and timely mechanical support/transplant to prevent complications.
- **Reproductive options:** prenatal diagnosis and preimplantation genetic testing (PGT-M) where familial variants are known; partner carrier screening.
- **Immunization/public-health/environmental measures:** not applicable (monogenic disease).

14. Other Species / Natural Disease

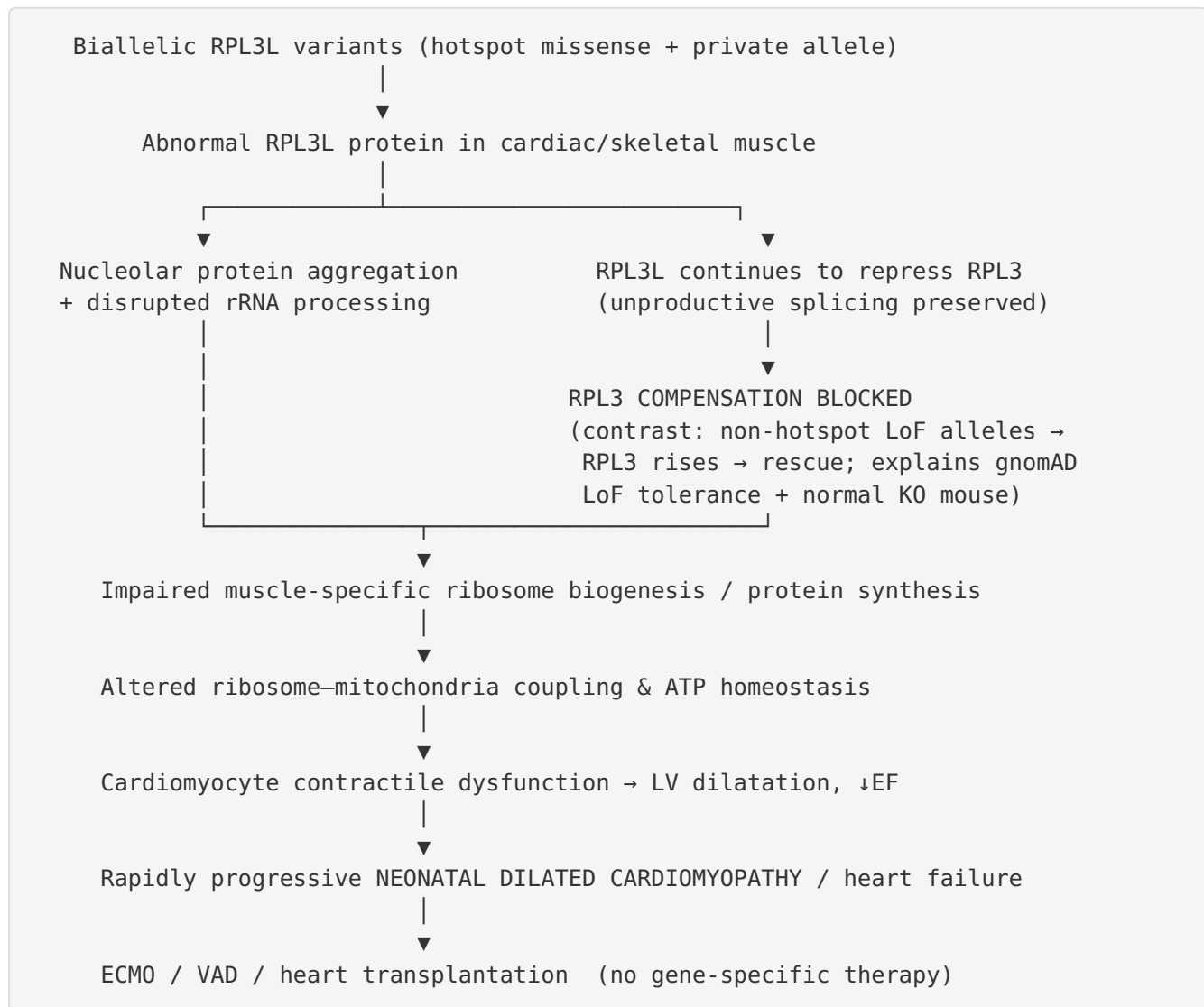
- **Taxonomy / orthologs:** *RPL3L* conserved in vertebrates (*Homo sapiens*, NCBI Taxon 9606). Mouse ortholog **Rpl3l** (NCBI Gene 66211; MGI:1913461; ENSMUSG00000002500; chromosome 17).

- **Comparative biology:** *Rpl3l*-knockout mice are viable with no overt cardiac phenotype, owing to compensatory RPL3 upregulation [PMID: 36882085](#) — a key species difference from the human recessive-lethal phenotype, explained by the GoF/compensation-blocking mechanism of human hotspot alleles [PMID: 41495453](#).
- **Natural disease/zoonosis:** no naturally occurring RPL3L cardiomyopathy documented in companion animals/wildlife; not zoonotic. (Naturally occurring canine DCM exists but is not RPL3L-linked in the literature.)

15. Model Organisms

- **Mouse:** *Rpl3l*-knockout — viable, no overt cardiac phenotype (RPL3 compensation; increased ribosome-mitochondria interaction and ATP); models the **LoF/compensated** branch but **does not recapitulate** human heart failure [PMID: 36882085](#). This limitation is itself mechanistically informative.
 - **Isogenic human cell / iPSC-cardiomyocyte models:** demonstrate hotspot-variant nucleolar aggregation, rRNA-processing disruption, and blocked RPL3 compensation — the best system reproducing the GoF mechanism [PMID: 41495453](#).
 - **Applications:** dissecting ribosome specialization, RPL3/RPL3L autoregulation, allele-selective therapeutics.
 - **Limitations:** rodent compensation masks the phenotype; humanized/hotspot-knock-in models would better mirror disease.
 - **Resources:** MGI:1913461 (mouse *Rpl3l*), Alliance of Genome Resources; patient-derived iPSC lines.
-

Mechanistic Model / Interpretation



The central, unifying insight is the **paralog-compensation switch**. *RPL3L* loss alone is well tolerated (population LoF tolerance; asymptomatic KO mouse) because *RPL3* rises to substitute. CMD2H arises specifically when an allele **both disables *RPL3L* and prevents *RPL3* rescue** — reconciling the genetic (recessive), population-genetic (LoF-tolerant), model-organism (KO-normal), and molecular (nucleolar aggregation) observations into one coherent narrative. This has direct therapeutic implications: strategies that **restore or permit *RPL3* upregulation** (allele-selective silencing of the toxic hotspot allele, or de-repression of *RPL3*) are the most mechanistically rational future interventions.

Evidence Base

PMID	Title (abridged)	Role in this report
32514796	<i>Bi-allelic missense variants in RPL3L associate neonatal DCM with muscle-specific ribosome biogenesis</i>	Establishes <i>RPL3L</i> as causal gene (5 individuals/3 families)
35323613	<i>Further evidence of autosomal recessive inheritance</i>	Independent confirmation; rapidly progressive, poor prognosis
41495453	Hotspot-variant mechanism	Nucleolar aggregation, disrupted rRNA processing, blocked RPL3 compensation
39803500	<i>Pathogenetic mechanisms of muscle-specific ribosomes in DCM</i>	Mechanistic review of muscle-specific ribosome role
36882085	<i>RPL3/RPL3L interplay modulates mitochondrial activity</i>	Compensation biology; ribosome–mitochondria/ATP; KO mouse
30271950	Coding variants and atrial fibrillation	Links <i>RPL3L</i> to cardiac electrophysiology (AF)
32822252	Genetic determinants of P-wave duration	<i>RPL3L</i> among atrial-conduction/AF loci
42360425	National pediatric cardiomyopathy cohort	Incidence 0.9/100,000 children/year
38427064	ESC EURObservational registry	DCM = 32.5% of pediatric cardiomyopathy; genetics/outcomes
38722325	NT-proBNP prognosis	Validates NT-proBNP prognostic biomarker
42334151	HeartMate 3 vs transplant in DCM	Modern support/transplant survival context
18652581	Cardiomyopathy in newborns/infants	Neonatal/infantile DCM natural history/prognosis
31298178	Pediatric HF outcomes in VAD era	Support/transplant/recovery outcomes
40645388	Barth syndrome/ <i>TAFazzin</i> case	Differential diagnosis (metabolic neonatal DCM)

PMID	Title (abridged)	Role in this report
30384889	Genetic basis of severe childhood cardiomyopathy	Differential/genetic landscape
32458740	Genetic testing in pediatric DCM	Diagnostic yield context
35849328	Cerebrovascular events on Berlin Heart EXCOR	Complications of neonatal mechanical support
38053305	Left-heart decompression on ECMO	Interventional management context
42159538	Sex/age genetic risk in DCM/ACM (SHaRe)	Population sex/age context (not <i>RPL3L</i> -specific)

Evidence-source classification. Human clinical/genetic: PMIDs 32514796, 35323613, 41495453 (also isogenic in vitro), 38427064, 42360425, 38722325, 42334151, 30271950, 32822252, 18652581, 31298178. Model organism: 36882085. Review/synthesis: 39803500. Population/computational: gnomAD, ClinVar, OMIM/MONDO/OLS4.

Limitations and Knowledge Gaps

- 1. Small clinical evidence base.** The gene–disease relationship rests on a handful of families/individuals. Penetrance, expressivity, full phenotypic spectrum, and any extracardiac (skeletal-muscle) manifestations are imprecisely defined.
- 2. No CMD2H-specific epidemiology.** Prevalence, incidence, carrier frequency, and any founder/geographic distribution of the hotspot alleles are unquantified; epidemiologic figures here are borrowed from the broader pediatric DCM population.
- 3. Mechanistic inferences.** Steps linking blocked RPL3 compensation → energetics → contractile failure are partly inferred from paralog biology and mouse data; direct in-human demonstration of the downstream cardiomyocyte energetic defect is incomplete.
- 4. Imperfect animal model.** The KO mouse does not recapitulate disease (compensation); a knock-in of a human hotspot allele would be needed to faithfully model the gain of function.
- 5. Variant interpretation.** Many *RPL3L* variants are missense with mechanism-dependent pathogenicity (compensation-blocking vs not); standard ACMG criteria may under-call the

crucial hotspot alleles without functional assays, and may over-call isolated truncating alleles given LoF tolerance.

6. **No targeted therapy or trials.** No CMD2H-specific pharmacologic, gene, or RNA therapies exist; treatment evidence is extrapolated from general pediatric DCM management.

Proposed Follow-up Experiments / Actions

1. **Generate a hotspot knock-in mouse (or iPSC-cardiomyocyte) model** to test whether a human *RPL3L* hotspot allele recapitulates DCM and to validate the compensation-block mechanism in vivo.
 2. **Systematic functional assay** (nucleolar aggregation, rRNA processing, RPL3 protein levels) for all reported/novel *RPL3L* missense variants, to classify compensation-blocking vs LoF and improve ACMG interpretation.
 3. **Build a genotype–phenotype registry** aggregating all CMD2H cases to define penetrance, natural history, biomarker trajectories (NT-proBNP), and outcomes.
 4. **Therapeutic proof-of-concept:** test allele-selective ASO/siRNA silencing of the toxic hotspot allele, or RPL3 de-repression/augmentation, in patient-derived cardiomyocytes.
 5. **Skeletal-muscle phenotyping** of affected individuals to determine whether a subclinical myopathy accompanies the cardiomyopathy.
 6. **Estimate carrier frequency** of the two hotspot alleles across gnomAD/population datasets to inform carrier screening and reproductive counseling.
 7. **Clarify the AF connection:** assess whether CMD2H probands or heterozygous carriers show measurable atrial-conduction abnormalities, integrating population AF/P-wave signals with the Mendelian disease.
-

Report compiled from 7 confirmed findings and 25 reviewed papers over 5 investigation iterations. Ontology annotations: MONDO:0859358; HGNC:10351; GO:0002181/0006412/0006941/0003735/0022625/0005730; CL:0000746; UBERON:0000948/0002084; NCIT:C15325/C50076. Evidence spans human clinical genetics, isogenic in-vitro models, mouse models, human GWAS, and pediatric DCM registries.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery